Johnson & Johnson Time Series Analysis

A reproducible analysis with R and Quarto

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1 Introduction

The Johnson & Johnson quarterly earnings per share series (1960–1980) is a canonical example in the time series literature, appearing in Shumway and Stoffer's *Time Series Analysis and Its Applications* as the first worked example of regression with seasonal dummies.

The series consists of 84 quarterly observations of earnings per share, starting in 1960-Q1 and ending in 1980-Q4. It exhibits a strong upward trend, clear quarterly seasonality, and cyclic structure at business-cycle frequencies. These features make it an excellent case study for comparing classical regression-based approaches with modern time series models.

1.1 Objective

This report provides a reproducible analysis of the J&J series using three complementary approaches:

1. **Regression baseline.** A log-linear model with a centered time trend and quarterly dummy variables, following the original exercise.
2. **Exploratory decomposition.** STL decomposition, seasonal plots, and quarterly boxplots to characterize the structure of the series.
3. **Alternative models and forecasting.** SARIMA and ETS models, validated via holdout and rolling cross-validation.

The analysis is fully reproducible: all figures, tables, and numerical results are generated from code in this repository.

2 Data

The dataset used in this report is the `JohnsonJohnson` series shipped with base R. It contains 84 quarterly observations, starting in 1960-Q1 and ending in 1980-Q4.

Earnings per share (EPS) is a standard financial metric computed by dividing a company's net income by the number of outstanding shares. It measures how much profit the company generated for each share of stock during a given period.

For Johnson & Johnson, EPS is reported quarterly, and this dataset tracks it from the first quarter of 1960 through the fourth quarter of 1980.

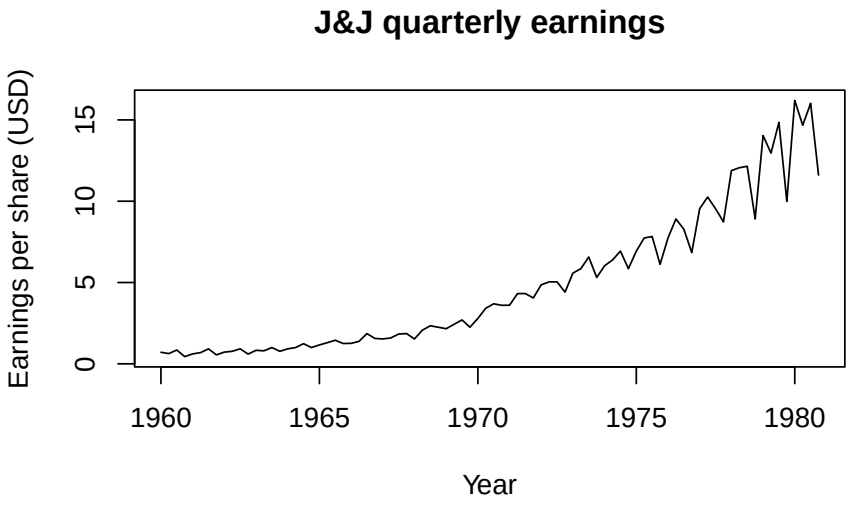


Figure 1: Johnson & Johnson quarterly earnings per share, 1960–1980.

As shown in Figure 1, the series displays two features that shape the modeling strategy.

First, an upward trend: earnings per share roughly triple over the twenty-one years.

Second, a clear seasonal oscillation: within each year, earnings rise from the first to the third quarter and drop in the fourth. This pattern repeats with remarkable consistency across the sample.

The variance also appears to grow with the level. Early years fluctuate within a narrow band, while later years show much larger swings. This pattern, called heteroscedasticity, motivates the log transformation used throughout the analysis.

All subsequent work is performed on the log series, $x_t = \log(y_t)$, where y_t denotes the original earnings per share.

3 Exploratory analysis

Before fitting any model, we inspect the structure of the series. We use the STL decomposition to separate trend and seasonality, quarterly boxplots to check whether the seasonal pattern is stable across years, and the sample ACF and PACF to guide model selection.

3.1 STL decomposition

STL (Seasonal-Trend decomposition using LOESS) splits a series into three additive components: a smooth trend, a repeating seasonal pattern, and a remainder that captures everything else. It is a data-driven alternative to fitting a regression first.

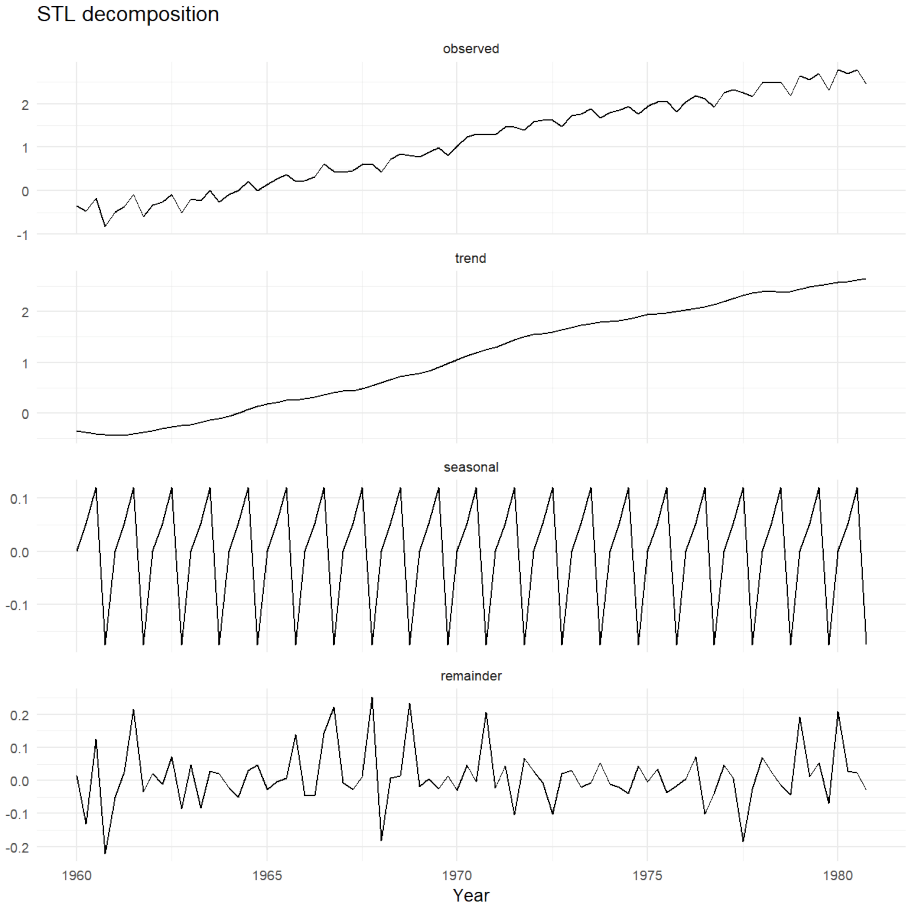


Figure 2: STL decomposition of log earnings. Panels show observed, trend, seasonal, and remainder components.

Figure 2 shows four panels. The observed panel reproduces the raw series, growing across the year and dropping in Q4, following the same pattern seen in the boxplots. The trend panel is a smoothed version of the series, with amplitude much smaller than the raw data. The seasonal component is approximately periodic, oscillating between about plus and minus one-tenth of a log unit. The remainder stays within a small band around zero, roughly two-tenths of a unit on the log scale, indicating that trend and seasonality capture most of the variation in the series.

3.2 Seasonal pattern

A seasonal plot rearranges the series into a grid with one row per year and one column per quarter. Each line traces the path of a single year across the four quarters, and the color of the line encodes the year, from grey (1960) to red (1980).

This layout makes the within-year pattern visible at a glance. If earnings follow a stable seasonal shape, all lines rise and fall the same way, and only their vertical position changes from year to year. Deviations from the common shape stand out immediately.

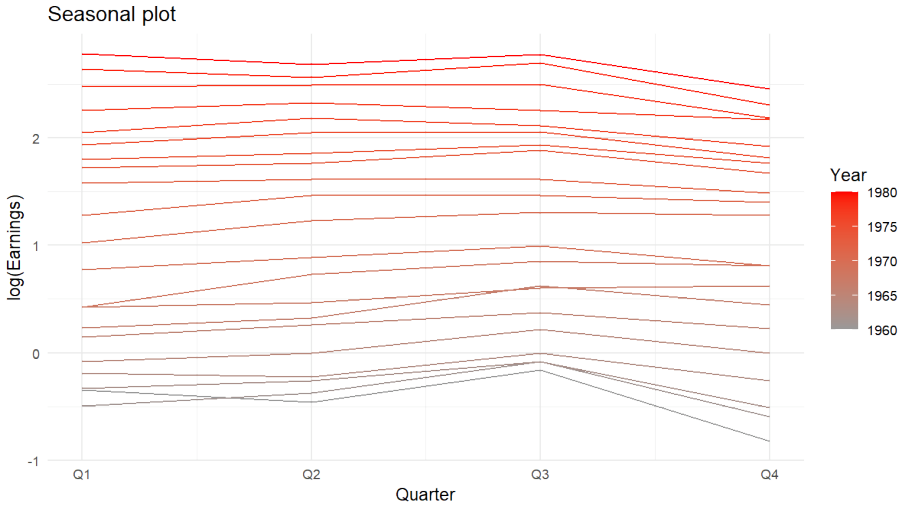


Figure 3: Seasonal plot of log earnings. Each line represents one year; color encodes year from grey (1960) to red (1980).

Figure 3 confirms a consistent within-year pattern: earnings grow from Q1 to Q3 and decline in Q4. The color gradient shows that later years (red) are systematically above earlier years (grey), reflecting the upward trend. No year deviates visibly from this shape, which justifies a fixed seasonal specification.

3.3 Quarterly distribution

A boxplot summarizes the distribution of a set of values in a compact form. The central box spans the interquartile range, from the 25th to the 75th percentile. The thick horizontal line inside the box marks the median. The whiskers extend to the most extreme values within 1.5 times the interquartile range, and points beyond them are outliers.

Here we group the 84 log earnings values by quarter. Each of the four boxplots therefore summarizes 21 observations, one per year.

Figure 4 shows that the median increases from Q1 to Q3 and Q4 presents larger dispersion. Q3 exhibits the narrowest interquartile range, indicating the least dispersion of log earnings in that quarter. If we rearrange the quarters to place Q4 before Q1, a pattern emerges that is easy to miss in the standard order: Q4

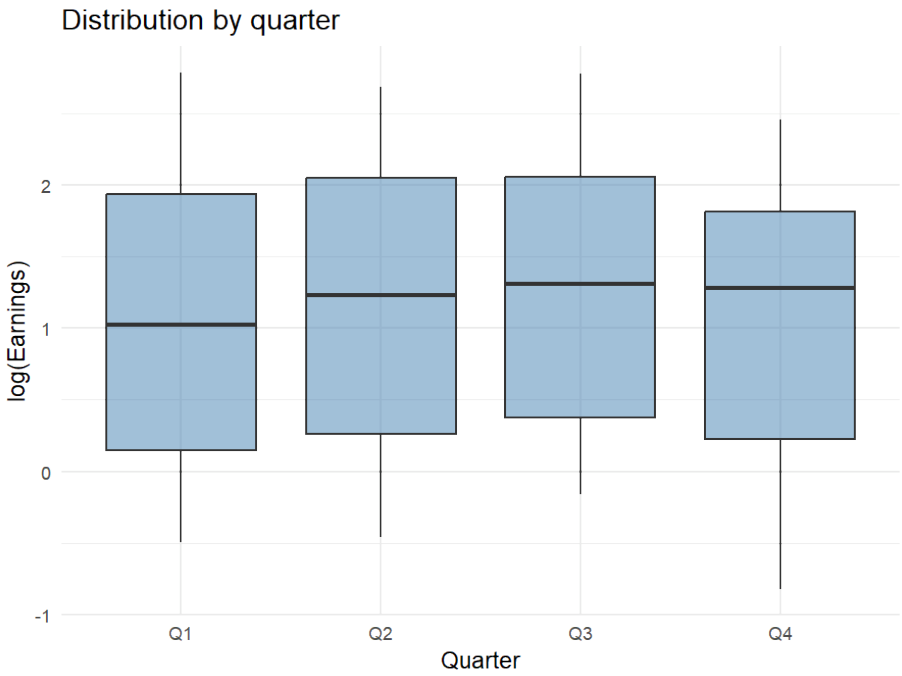


Figure 4: Distribution of log earnings by quarter across the 21 years.

is systematically above Q1, Q1 is the absolute minimum, and Q3 is the absolute maximum. The within-year trajectory is therefore not monotonic but cyclical, with a pronounced drop between Q3 and Q4 followed by a further drop into Q1, and then a recovery through Q2 and Q3.

3.4 ACF and PACF

The sample autocorrelation function (ACF) measures the linear correlation between the series and its own lagged values. The partial autocorrelation function (PACF) does the same but removes the effect of intermediate lags. Together they are the standard diagnostic for choosing the orders of an ARIMA model.

Figure 5 shows that the ACF decays very slowly, which is the classic signature of a non-stationary series. The PACF, by contrast, has a single significant spike at lag 1. Beyond lag 1, it shows local minima at lags 5, 9, and 13, all of which stay within the confidence band. The isolated spike at lag 1 suggests a first-order autoregressive structure, while the regular spacing of the local minima hints at a seasonal component with a shift of one quarter. The slow decay of the ACF and the shifted minima together motivate the seasonal differencing used in the modeling section.

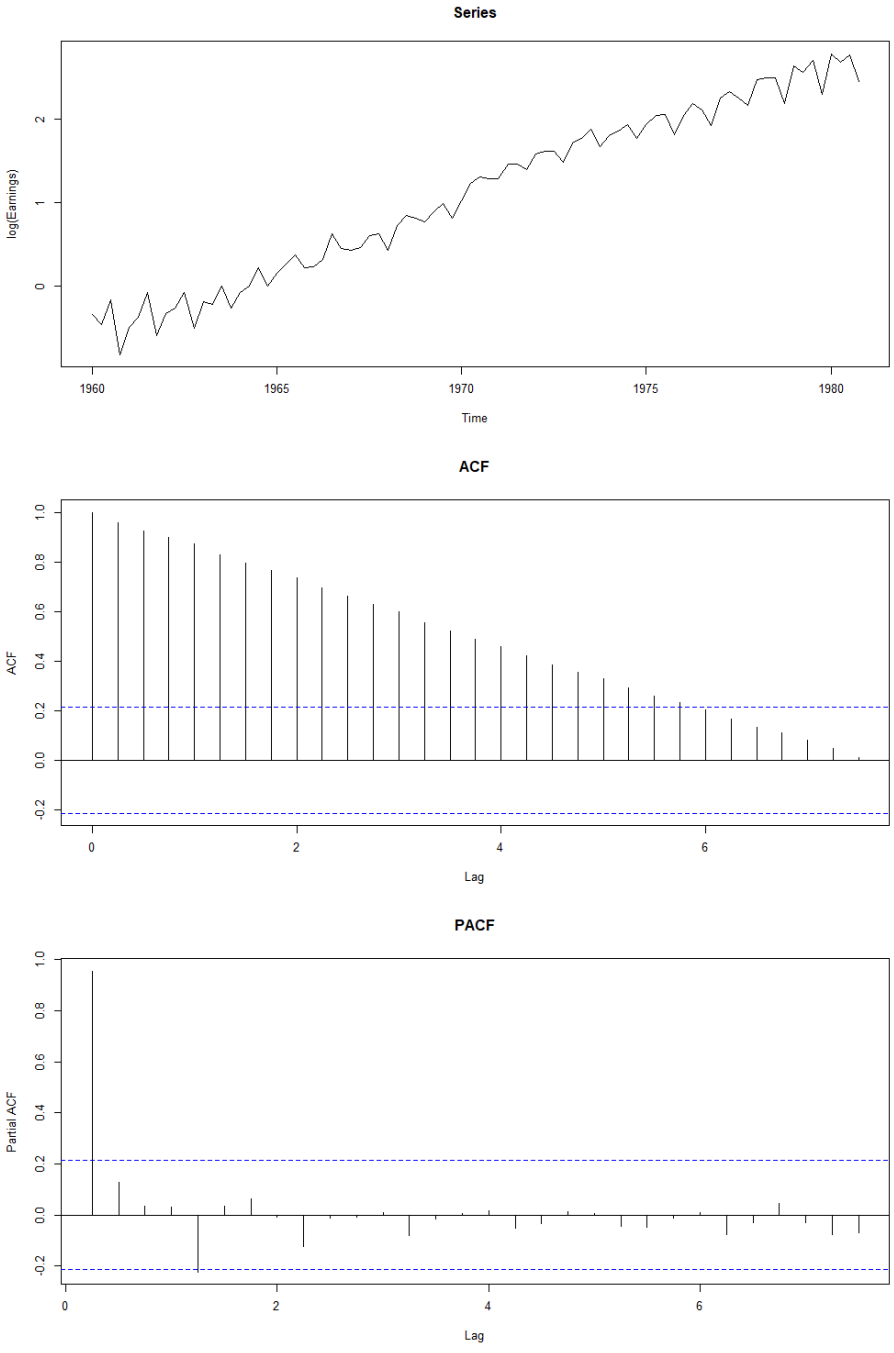


Figure 5: Time series (top), sample ACF (middle), and sample PACF (bottom) of log earnings.

4 Regression baseline

The original exercise proposes a log-linear model with a centered time trend and quarterly dummy variables:

$$x_t = \beta t + \alpha_1 Q_1(t) + \alpha_2 Q_2(t) + \alpha_3 Q_3(t) + \alpha_4 Q_4(t) + \omega_t$$

where $x_t = \log(y_t)$, t is the time index centered at 1970, $Q_i(t) = 1$ if observation t belongs to quarter i and 0 otherwise, and ω_t is a Gaussian white noise sequence. The model has no separate intercept because the four quarterly dummies span the constant.

The parameters have a direct interpretation. The trend coefficient β captures the average quarterly growth rate of log earnings across the entire sample. Each α_i represents the baseline level of quarter i after removing the trend. The seasonal amplitude is encoded in the differences $\alpha_i - \alpha_j$ between quarters.

4.1 Coefficient estimates

The table below reports the estimated coefficients, their standard errors, the test statistic, and the associated p-value for the no-intercept specification. Each coefficient is tested against zero: a small p-value indicates that the coefficient is statistically different from zero.

Table 1: Coefficient estimates for the no-intercept regression.

term	estimate	std.error	statistic	p.value
trend	0.1672	0.0023	73.9990	0
quarterQ1	1.0528	0.0274	38.4804	0
quarterQ2	1.0809	0.0274	39.4999	0
quarterQ3	1.1510	0.0274	42.0350	0
quarterQ4	0.8823	0.0274	32.1858	0

Table 1 shows the estimated coefficients. The trend estimate $\hat{\beta}$ is positive and highly significant, confirming the upward movement visible in the series. All four quarterly coefficients are also significant at conventional levels. They differ from each other, and their spread reflects the amplitude of the seasonal pattern identified in the exploratory analysis.

4.2 Observed vs fitted

Overlaying fitted values on the observed series is the quickest way to judge how well a model tracks the data. Two failure modes are common: a model that captures the trend but misses the seasonality, or one that captures the seasonality but misses long-run movements.

Figure 6 shows that the regression captures the overall upward trend and the quarterly pattern, but the fitted values form a smooth quasi-periodic curve that misses the year-to-year variation in the observed series. This is expected: a

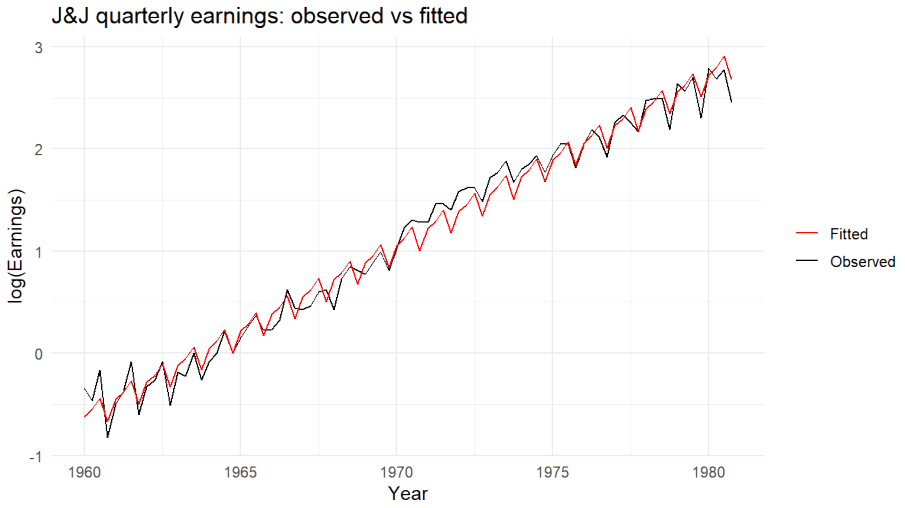


Figure 6: Observed log earnings (black) and fitted values from the no-intercept regression (red).

regression with five parameters cannot capture business-cycle dynamics. The residuals, the difference between the two curves, are examined next.

4.3 Residual diagnostics

Residuals are the differences between observed and fitted values. If the model captures all systematic structure, the residuals should behave like white noise: no trend, no seasonal pattern, no autocorrelation. Non-random residuals signal that something is missing from the specification.

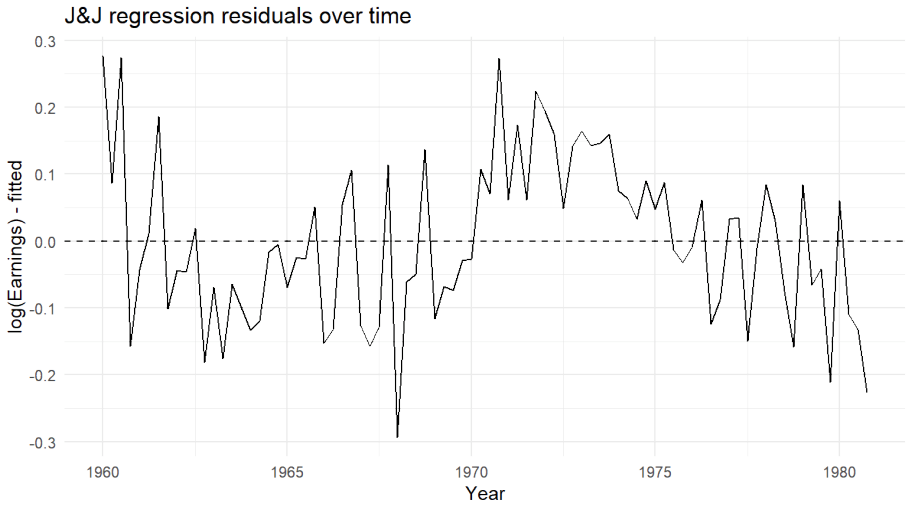


Figure 7: Residuals from the no-intercept regression.

The residuals in Figure 7 are not white noise. They display a low-frequency oscillation with a clear structure: a sustained negative phase through the 1960s, a sharp positive peak centered around 1972, and a return to negative values after 1976. The peak is robust across moving-average windows of different lengths, which confirms it is a genuine feature of the residuals rather than an artifact of smoothing. The oscillation is not strictly periodic and its duration is not fixed, which is characteristic of economic cycles rather than of a mechanical seasonal pattern. The regression specification, with only five parameters and a rigid trend, cannot represent this structure, motivating the use of autoregressive models such as SARIMA in the next section.

5 Stationarity

Most time series models, including SARIMA and ETS, assume the series is stationary after appropriate transformations. A series is weakly stationary if its mean, variance, and autocovariance structure do not change over time. We test this assumption with two complementary stationarity tests, and we also apply the Ljung-Box test to check whether the series exhibits serial dependence.

5.1 ADF and KPSS

The Augmented Dickey-Fuller (ADF) test has the null hypothesis that the series contains a unit root, meaning it is non-stationary. The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test has the opposite null: that the series is stationary. Using both tests together reduces the risk of a misleading conclusion.

Table 2: Stationarity tests on the log series and its first difference.

series	test	statistic	p_value	null_hypothesis	conclusion
log series	ADF	-1.1543	0.9087	Unit root (non-stationary)	Non-stationary
log series	KPSS	2.1867	0.0100	Level-stationary	Non-stationary
first difference	ADF	-4.5649	0.0100	Unit root (non-stationary)	Stationary
first difference	KPSS	0.1720	0.1000	Level-stationary	Stationary

Table 2 shows that both tests agree on the log series: ADF fails to reject the unit-root null, and KPSS rejects stationarity. On the first difference, both tests agree in the opposite direction. The conclusion is that the log series is non-stationary and a single regular difference is enough to make it stationary.

5.2 Ljung-Box test

The Ljung-Box test checks whether a series exhibits significant autocorrelation up to a given lag. Applied to the log series, it confirms the visual impression from the ACF.

Table 3: Ljung-Box test on the log series, lag = 10.

statistic	df	p_value
613.0387	10	0

The Ljung-Box test checks whether a series shows autocorrelation up to a given lag. Its null hypothesis is that the values are independent over time. The test statistic of 613 with 10 degrees of freedom produces a p-value below any conventional threshold. The null is rejected decisively: the log series has strong serial dependence, and its values are far from independent. This provides one more reason to move beyond a regression with independent errors toward a model that explicitly captures the autocorrelation, such as SARIMA.

6 Alternative models

The regression baseline captures trend and seasonality but leaves a cyclic residual structure unexplained. We now fit two families of models specifically designed to handle autocorrelation: SARIMA and ETS.

6.1 Model comparison

How should we choose between SARIMA and ETS? Both model the same series, but they represent different assumptions about the underlying structure. Information criteria provide a principled comparison. AIC and BIC trade off goodness of fit against model complexity: lower values indicate a better balance. They are not directly comparable across different datasets, but they are meaningful when comparing models fitted to the same data.

Table 4: Information criteria for the SARIMA and ETS fits on the full log series.

model	log_lik	aic	bic
SARIMA	82.47	-154.95	-143.04
ETS	19.28	-20.56	1.32

Table 4 compares the two models using log-likelihood, AIC, and BIC. The SARIMA model has an AIC of -154.95, about 134.4 points lower than ETS. The BIC difference follows the same direction and is even larger in magnitude. All three criteria therefore point to the SARIMA model as the better trade-off between fit and complexity.

6.2 SARIMA specification

SARIMA extends ARIMA to handle seasonal structure. Its full specification is ARIMA$(p, d, q)(P, D, Q)[s]$, where:

- p, d, q are the regular autoregressive, differencing, and moving-average orders.

- P, D, Q are the seasonal counterparts.
- s is the seasonal period, equal to 4 for quarterly data.

We fit the model automatically with `forecast::auto.arima`, which searches over a grid of orders and selects the one minimizing AICc.

The search selects a SARIMA(2,0,0)(1,1,0)[4] with drift. The two regular autoregressive terms capture short-run dependence between consecutive quarters. The seasonal term at lag 4 captures the annual cycle. The drift term reflects the average quarterly growth of log earnings. Because the seasonal differencing order is $D = 1$, the model effectively works with year-over-year changes, which removes the seasonal non-stationarity detected earlier.

6.3 ETS specification

ETS stands for *Error, Trend, Seasonal*. Each component can take different forms: error can be additive (A) or multiplicative (M); trend can be none (N), additive (A), or damped (Ad); seasonality can be none (N), additive (A), or multiplicative (M). The combination defines a family of exponential smoothing models.

We fit the model automatically with `forecast::ets`, which evaluates all feasible combinations and selects the one minimizing AICc.

The selected model is ETS(A,A,A), also known as Holt-Winters additive. It decomposes the series into level, trend, and seasonality, each updated exponentially with weights α, β, and γ. The estimated seasonal smoothing parameter is 0.54, indicating that the seasonal component adapts substantially each year.

7 Forecast validation

Comparing models on the same data used for fitting can be misleading. We evaluate out-of-sample performance with two complementary strategies: a single holdout split and rolling cross-validation.

7.1 Holdout validation

The series is split into a training set covering 1960 to 1975 (64 observations) and a test set covering 1976 to 1980 (20 observations). Both models are refit on the training set and evaluated on the test set.

Three metrics summarize the forecast errors:

- **RMSE** (Root Mean Squared Error) is the square root of the average squared error. It penalizes large errors more heavily than small ones and is on the same scale as the log earnings.
- **MAE** (Mean Absolute Error) is the average absolute error. It is less sensitive to outliers than RMSE.
- **MAPE** (Mean Absolute Percentage Error) expresses the error as a percentage of the observed value. It is useful for comparing across scales, but it is unstable when the true value is close to zero.

Table 5: Holdout accuracy for SARIMA and ETS on the 1976–1980 test set.

model	RMSE	MAE	MAPE
SARIMA	0.0833	0.0712	2.9650
ETS	0.2181	0.1815	7.1837

Table 5 shows that SARIMA achieves a test RMSE of 0.0833, a MAE of 0.0712, and a MAPE of 2.96 percent. ETS, in contrast, achieves an RMSE of 0.2181, a MAE of 0.1815, and a MAPE of 7.18 percent. The difference is substantial: SARIMA is roughly 2.6 times more accurate on the test set by RMSE.

7.2 Rolling cross-validation

Rolling cross-validation fits the model at each time point and predicts one step ahead, then averages the squared errors across all folds. This uses every observation as a potential test point.

Table 6: One-step-ahead rolling cross-validation RMSE.

model	rmse
SARIMA	0.1300
ETS	0.1389

Table 6 shows the same ranking: SARIMA has a lower cross-validated RMSE than ETS. Both metrics agree, giving confidence in the conclusion.

8 Discussion

Three findings stand out from the analysis.

First, the log transformation and the choice of a single difference are well justified by the data. The raw series shows variance growing with the level, and the log scale stabilizes it. The STL decomposition then reveals that trend and seasonality account for most of the variation, leaving a remainder with amplitude of roughly ± 0.2 on the log scale. The ADF and KPSS tests on the differenced series agree that a single regular difference suffices to achieve stationarity. This provides a clear justification for the `d = 1` setting that the modeling section assumes.

Second, the regression baseline is useful as an interpretable model but insufficient as a complete description of the series. All five coefficients are significant, and the fitted values track the trend and the quarterly pattern accurately. However, the residuals are not white noise: they display a low-frequency oscillation with irregular phases, including a sustained negative period through the 1960s, a sharp peak around 1972, and a return to negative values after 1976. The five-parameter specification cannot represent this structure. This is a genuine limitation of the model class, not a numerical artifact.

Third, the SARIMA model outperforms ETS on every metric considered. The automatic search selects a SARIMA(2,0,0)(1,1,0)[4] with drift, which combines regular autoregressive terms, a seasonal autoregressive term at lag 4, and seasonal differencing. The ETS model selects the additive Holt-Winters specification, ETS(A,A,A). SARIMA achieves an AIC of about 134 points lower than ETS, and a test-set RMSE that is roughly 2.6 times smaller. The rolling cross-validation confirms the same ranking. The most likely explanation is that the regular AR terms in SARIMA capture the short-run autocorrelation that the additive ETS specification cannot represent, while the seasonal AR term captures the annual cycle.

Two caveats apply to the third finding. First, the seasonal differencing order (`D = 1`) chosen by `auto.arima` differs from the regular differencing order (`d = 1`) suggested by the ADF and KPSS tests. Both approaches remove the non-stationarity, but they do so through different mechanisms. The AICc criterion preferred the seasonal-differencing solution for this series. Second, the holdout RMSE and the tsCV RMSE measure different things: the holdout evaluates a single model trained on 64 observations and asked to forecast 20 steps ahead, while the tsCV fits a new model at every step and predicts one step ahead. Both metrics agree on the ranking of the two models, which strengthens the conclusion.

9 Limitations

The analysis has several limitations worth noting.

The sample is small: 84 quarterly observations spanning 1960 to 1980. With only 21 years of data, the effective sample size for estimating seasonal and business-cycle structure is limited. Forecasts beyond 1980 should not be extrapolated without additional data, since the later behavior of the series may reflect economic conditions absent from this sample.

The comparison between SARIMA and ETS relies on automatic model selection. `auto.arima` and `ets` search over candidate specifications by AICc, and the selected models are the best within their respective families, but not necessarily the best possible models for this series. A manual search guided by the ACF and PACF could yield a different SARIMA specification. Similarly, the ranking of the two families could change under a different loss function.

The SARIMA and ETS models were compared on a single train/test split (1960–1975 for training, 1976–1980 for testing), and this split was also supplemented by rolling cross-validation. The single split places the test period in the last five years of the sample, when the series appears to flatten slightly. A different split, or a larger number of folds in the cross-validation, could change the magnitude of the reported differences, although the direction of the ranking was consistent across both evaluations.

The seasonal plot shows no year that deviates strongly from the overall shape, but this does not rule out structural breaks at a finer scale. A formal breakpoint analysis, such as the Chow test or the Bai–Perron procedure, was not performed. If the series contains a structural break, the models presented here would fit

neither regime well.

Finally, all figures and tables in this report are generated from a single pipeline. The pipeline is reproducible, but it assumes that the input series is `datasets::JohnsonJohnson` unchanged. Modifying the input or the transformation would require rerunning the analysis and updating the interpretation accordingly.

10 Reproducibility

All results in this report are generated from code in this repository. The analysis is organized into two stages.

The first stage runs the full pipeline: data loading, exploratory decomposition, regression fitting, stationarity tests, alternative models, and forecast validation. It writes all figures and tables to `output/` and saves the key R objects to `output/results.rds`.

Rscript analysis.R On systems where Rscript is not on the PATH, the same command can be run from within an R session opened at the project root: source("analysis.R") The second stage renders this report from the precomputed results. It does not refit any model; it reads output/results.rds and produces the final document. quarto render report.qmd To regenerate the PDF version with the LaTeX source preserved for external editing: quarto render report.qmd –to pdf The LaTeX source is written to report.tex alongside the PDF.

Package versions are pinned via renv. To restore the exact environment used to produce this report: renv::restore() The R session used to compile the report is recorded below. R version: R version 4.4.1 (2024-06-14 ucrt)

Key package versions:

```
Registered S3 method overwritten by 'quantmod':
  method            from
  as.zoo.data.frame zoo
```

forecast : 8.23.0 tseries : 0.10.58 ggplot2 : 3.5.1 broom : 1.0.7 knitr : 1.48 dplyr : 1.1.4 tibble : 3.2.1 rlang : 1.1.4 quarto : 1.5.1